



Fondamenti di Analisi dei Dati

from **data analysis** to **predictive techniques**

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Data Representation

Understanding how data transforms from tables to geometric points in feature spaces, and how representation functions unlock powerful machine learning capabilities.

From Data Tables to Feature Spaces

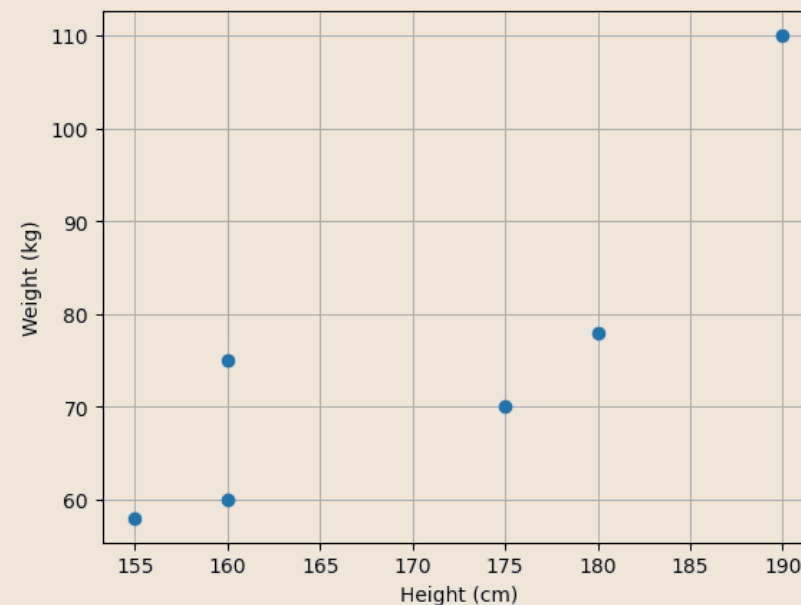
Traditional View

Data as tables with named variables and observations. Focus on interpreting individual variable values and their meanings.

When working with many variables, geometric interpretation becomes essential. For example, a 100×100 pixel image becomes a 10,000-dimensional vector where each dimension represents a specific pixel value.

Geometric View

Data as d-dimensional points in vector spaces. Variables become features, enabling mathematical operations and distance calculations.



Feature Extraction & Representation Functions

Given this interpretation of variables as features, it is often common to talk about **feature extraction**, in the case of any process transforming the data $\mathbf{x} \in \mathbb{R}^d$ to another form $\mathbf{y} \in \mathbb{R}^m$. In this context, a function:

$$f : \mathbb{R}^d \rightarrow \mathbb{R}^m$$

will be often referred to as a "**feature extraction function**" (of feature extractor) or a "**representation function**". We will also refer to the process of mapping data using a representation function as **data representation**.

We will also refer to $\mathbb{R}^d$ and $\mathbb{R}^m$ as "**feature spaces**" or "**representation spaces**" as they will be **vector spaces** to which data points (also called **feature vectors**) belong.

Understanding Norms: Measuring Vector Magnitude

A norm is a function that **maps vectors to non-negative real numbers, measuring distance from the origin**:

$$p : S \rightarrow \mathbb{R}_0^+$$

The L-p norm family provides different ways to quantify vector magnitude.

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$$

L2 Norm (Euclidean)

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$$

The standard "magnitude" of a vector
- the straight-line distance from origin to point.

L1 Norm (Manhattan)

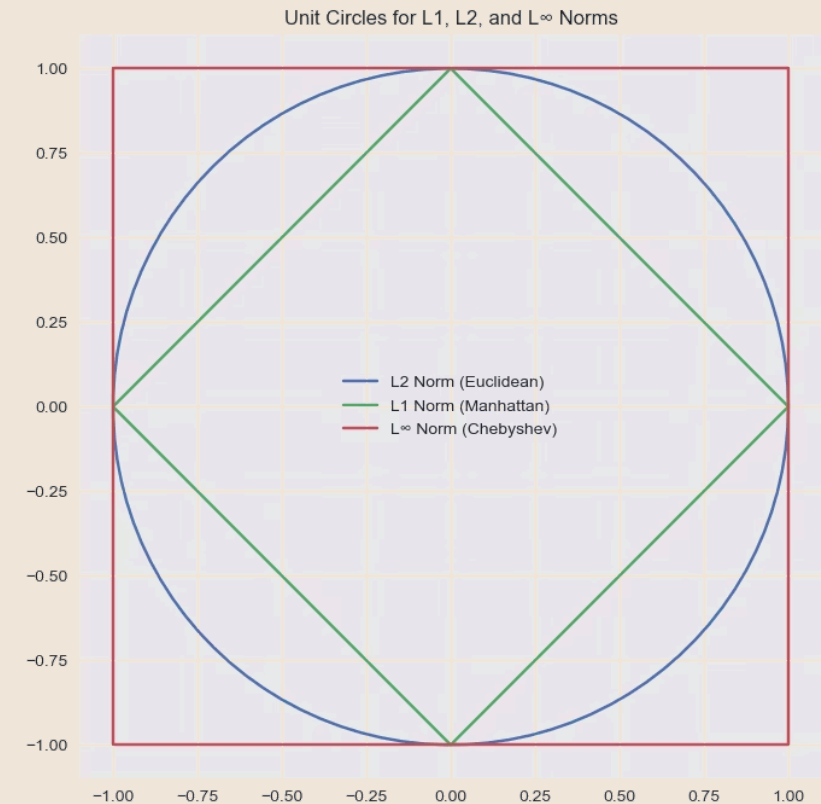
$$\|\mathbf{x}\|_1 = \sum_{i=1}^d |x_i|$$

Sum of absolute values of components. Named after city blocks in Manhattan.

L ∞ Norm (Maximum)

$$\|\mathbf{x}\|_\infty = \max\{|x_i|\}_{i=1}^d$$

The maximum absolute value among all vector components.



Metrics: Measuring Distance Between Points

A metric is a function

$$m : S \times S \rightarrow \mathbb{R}$$

that measures distance between points. Valid metrics must satisfy four key properties:

1. The distance of a point from itself is zero: $m(\mathbf{x}, \mathbf{x}) = 0$;
2. The distance between two distinct points is always positive (positivity) if $\mathbf{x} \neq \mathbf{y}$: $m(\mathbf{x}, \mathbf{y}) > 0, \forall \mathbf{x}, \mathbf{y} \in S, \mathbf{x} \neq \mathbf{y}$
3. The distance between x and y is the same as the distance between y and x (symmetry): $m(\mathbf{x}, \mathbf{y}) = m(\mathbf{y}, \mathbf{x})$.
4. Triangle inequality: $m(\mathbf{x}, \mathbf{y}) \leq m(\mathbf{x}, \mathbf{z}) + m(\mathbf{z}, \mathbf{y}), \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in S$

From the L-p norms, we can derive a **family of metrics** as follows:

$$L_p(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$$

1

L2 Distance (Euclidean)

$$L_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$$

Straight-line distance between two points in space.

2

L1 Distance (Manhattan)

$$L_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = \sum_{i=1}^d |x_i - y_i|$$

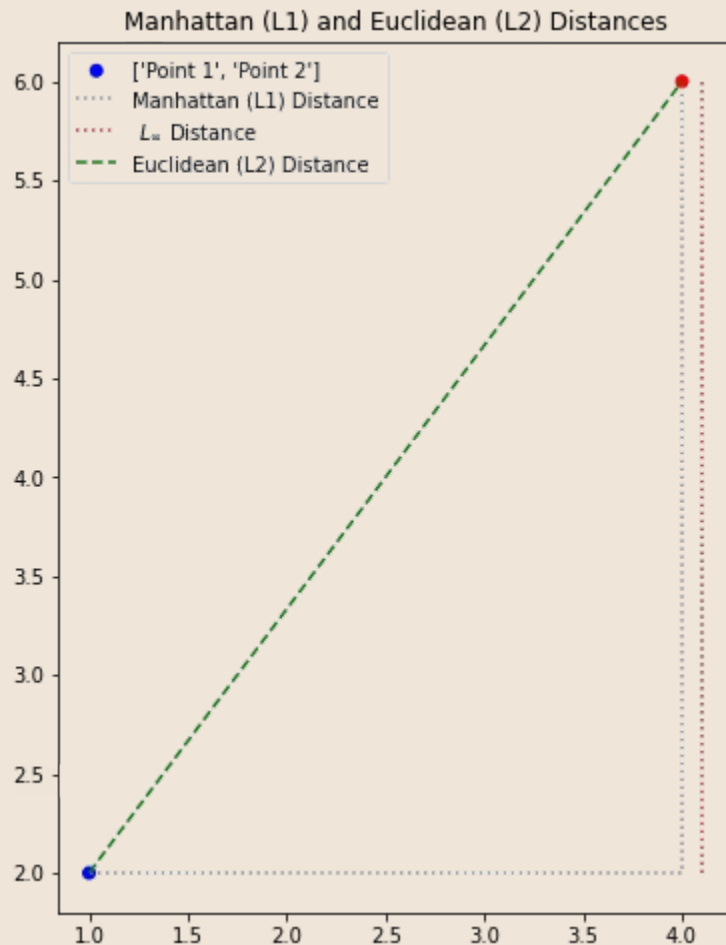
Distance measured along axis-aligned paths, like navigating city blocks.

3

L_∞ Distance

$$L_\infty(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_\infty = \max_{i=1}^d \{|x_i - y_i|\}$$

Maximum difference across any single dimension.



Manhattan vs. Euclidean Distance

Manhattan Distance

7.00 units

The path a taxi must drive along grid streets to reach destination.

Euclidean Distance

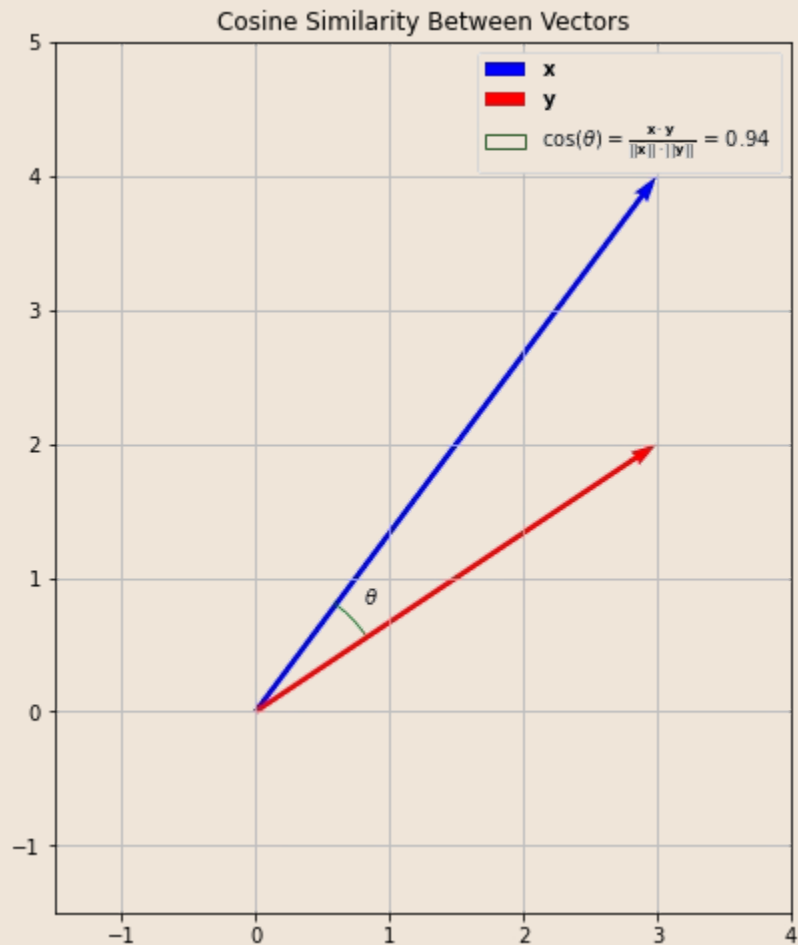
5.00 units

The straight-line "as the crow flies" distance between two points.

L-Inf Distance

4.00 units

The maximum coordinate difference between the two points.



Cosine Similarity: Measuring Orientation

Cosine similarity measures the angle between vectors, ignoring magnitude. This is crucial when scale doesn't matter - like comparing documents of different lengths where we care about word proportions, not total counts.

$$S_C(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

+1

Maximum Alignment

Vectors point in exactly the same direction - perfect similarity.

0

Orthogonal

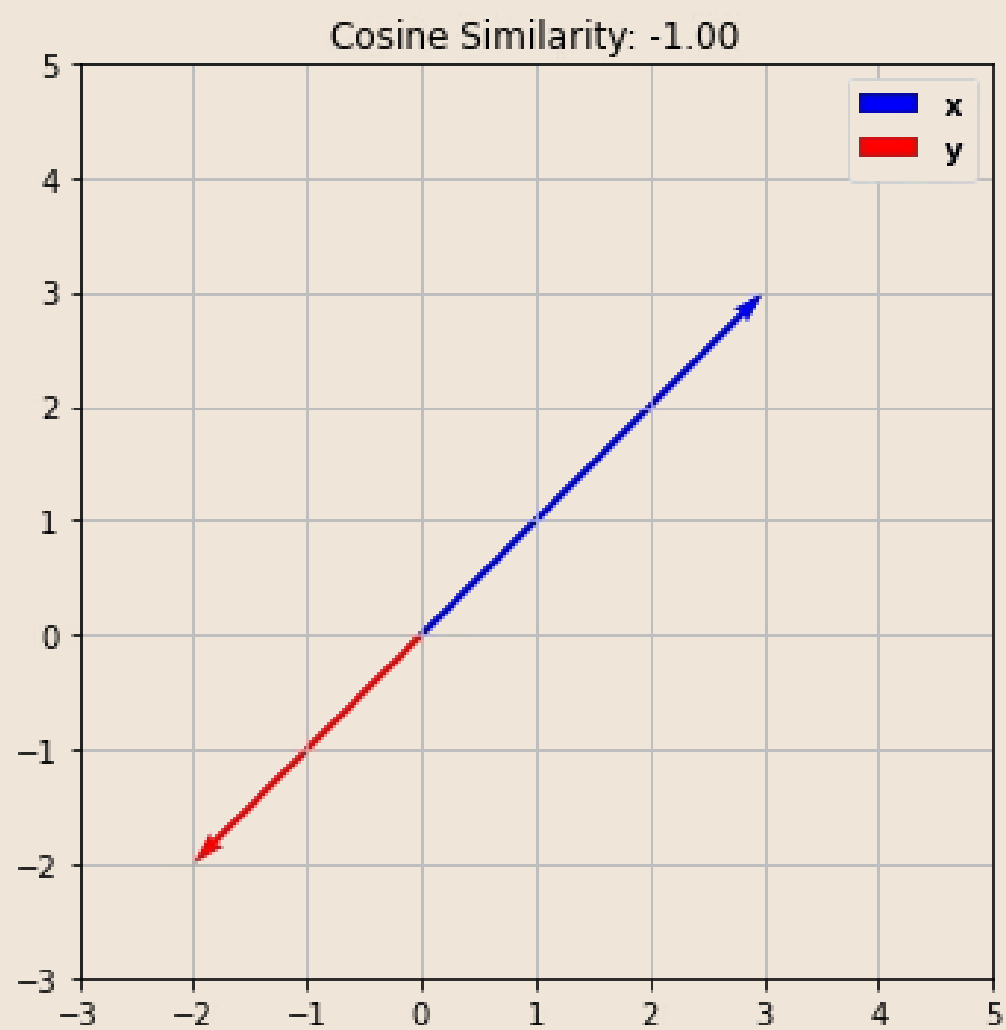
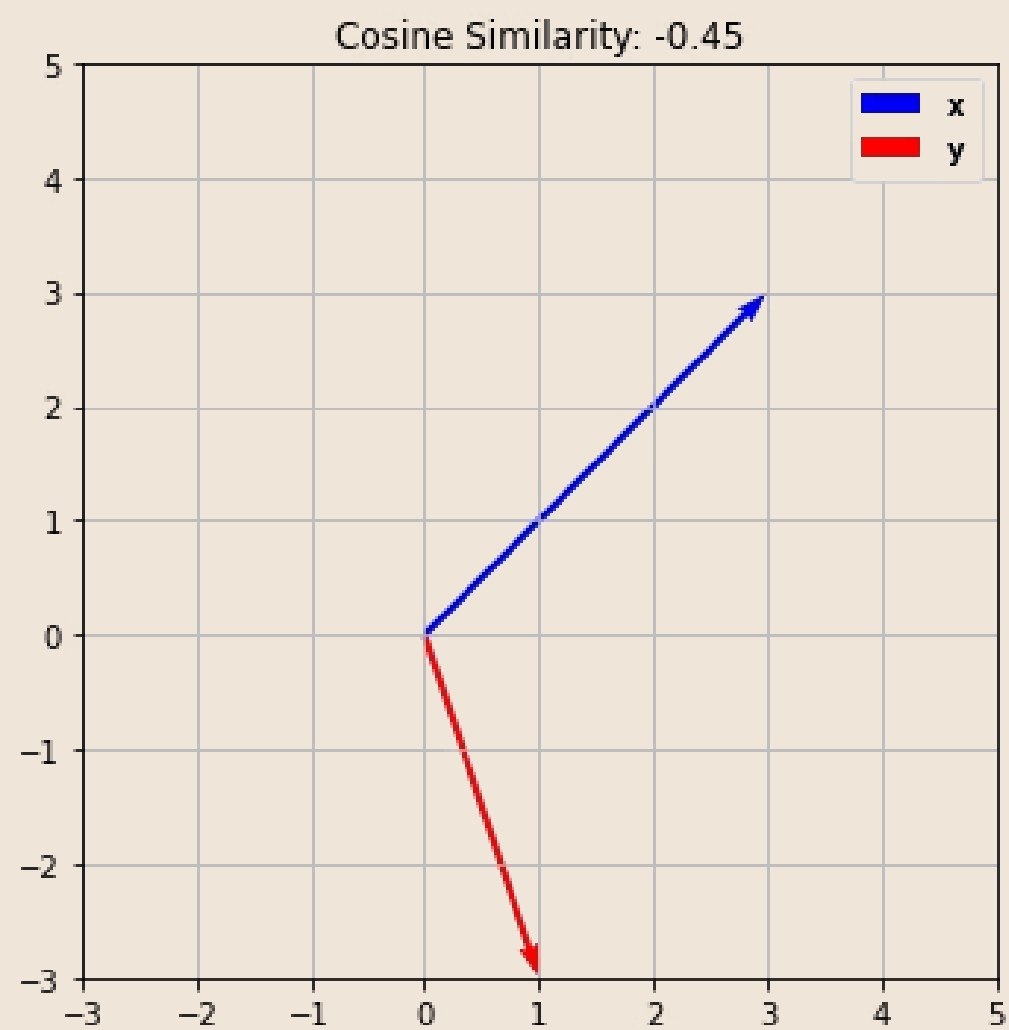
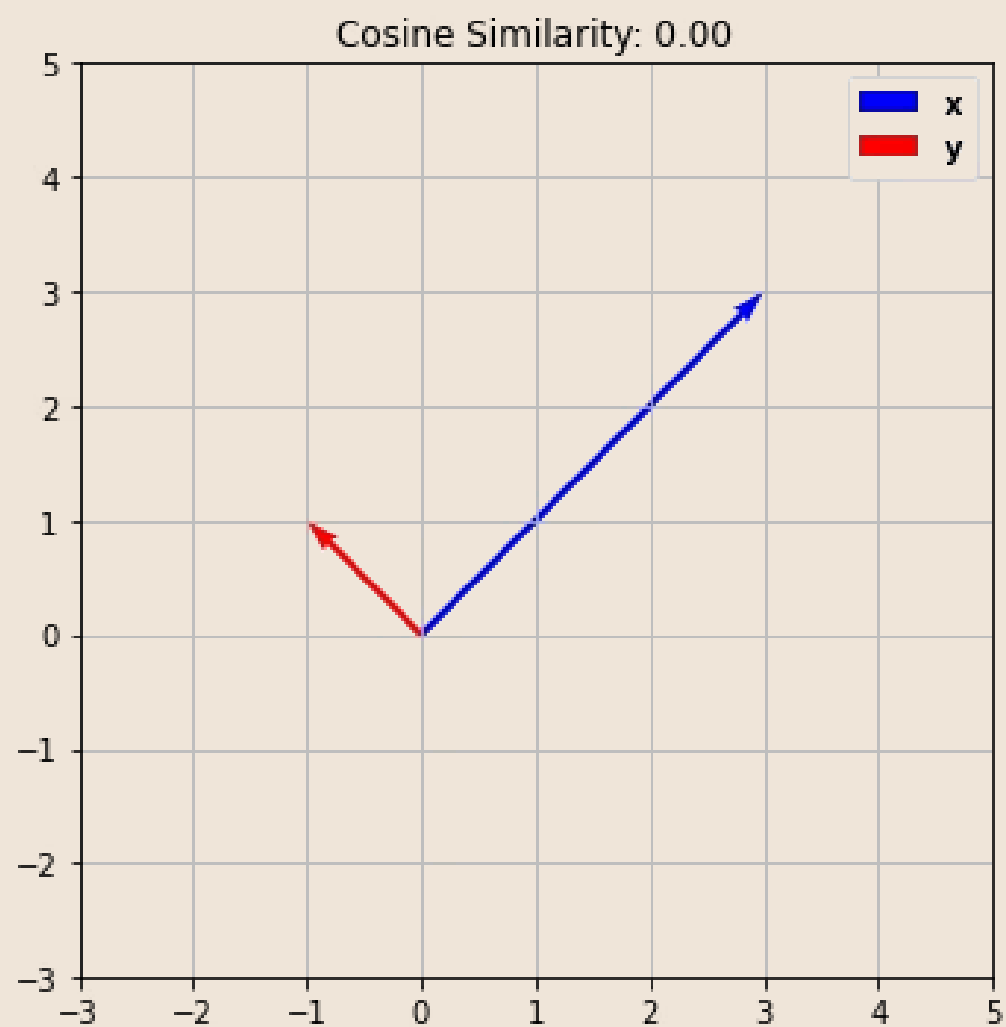
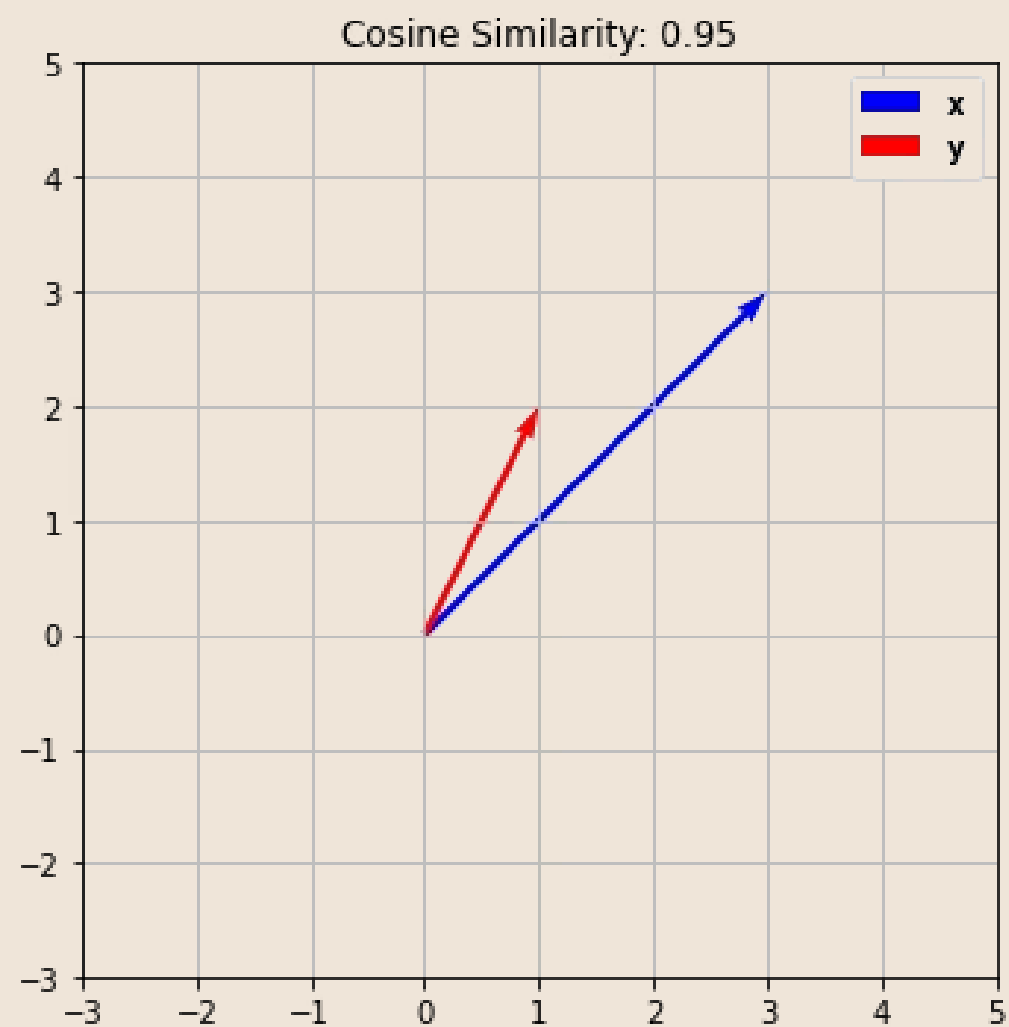
Vectors are perpendicular - no similarity or relationship.

-1

Opposite

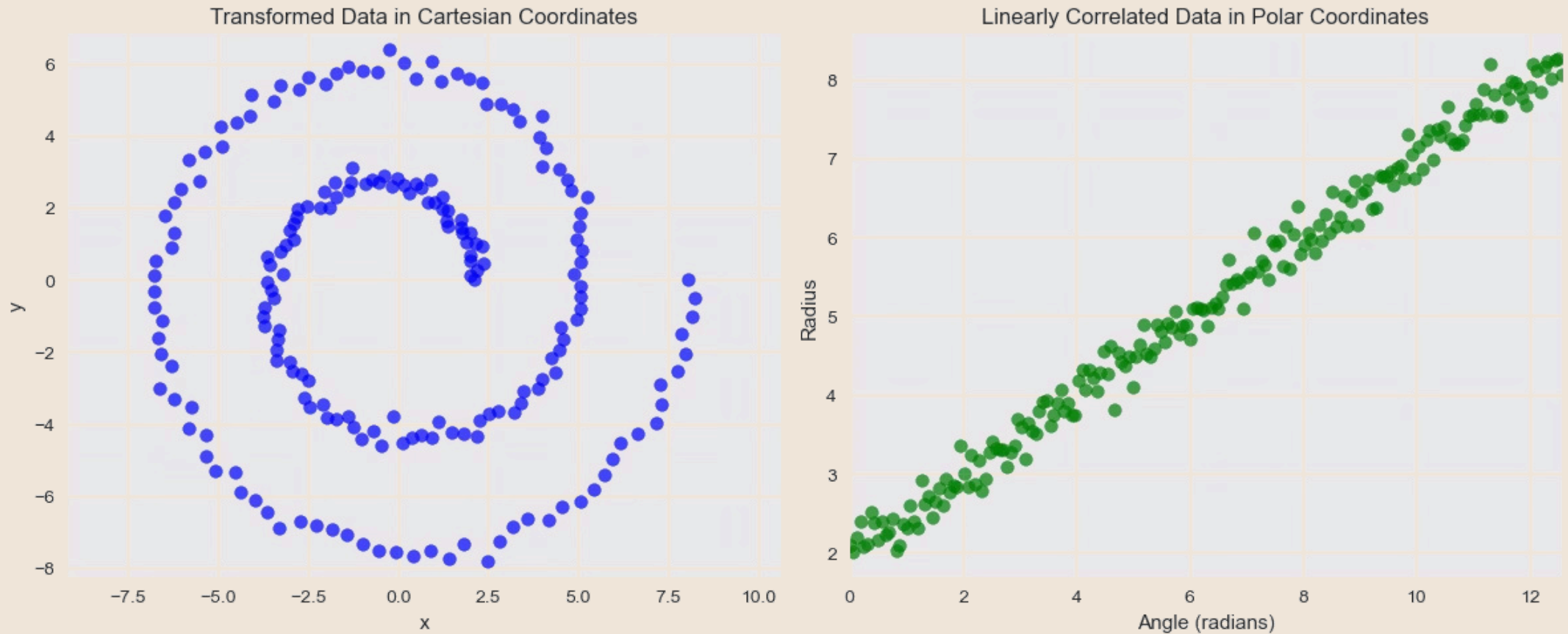
Vectors point in exactly opposite directions - maximum dissimilarity.

Examples of Cosine Similarity Values



The Power of Coordinate Transformation

The same data can appear completely different depending on the coordinate system. A complex spiral pattern in Cartesian coordinates becomes a simple linear relationship in polar coordinates.



This demonstrates a fundamental principle: choosing the right representation can transform an impossible problem into a trivial one. The transformation functions are:

Radius: $r = \sqrt{x^2 + y^2}$

Angle: $\theta = \arctan\left(\frac{y}{x}\right)$

What appears as a complex non-linear pattern becomes a simple linear relationship that can be modeled with basic regression.

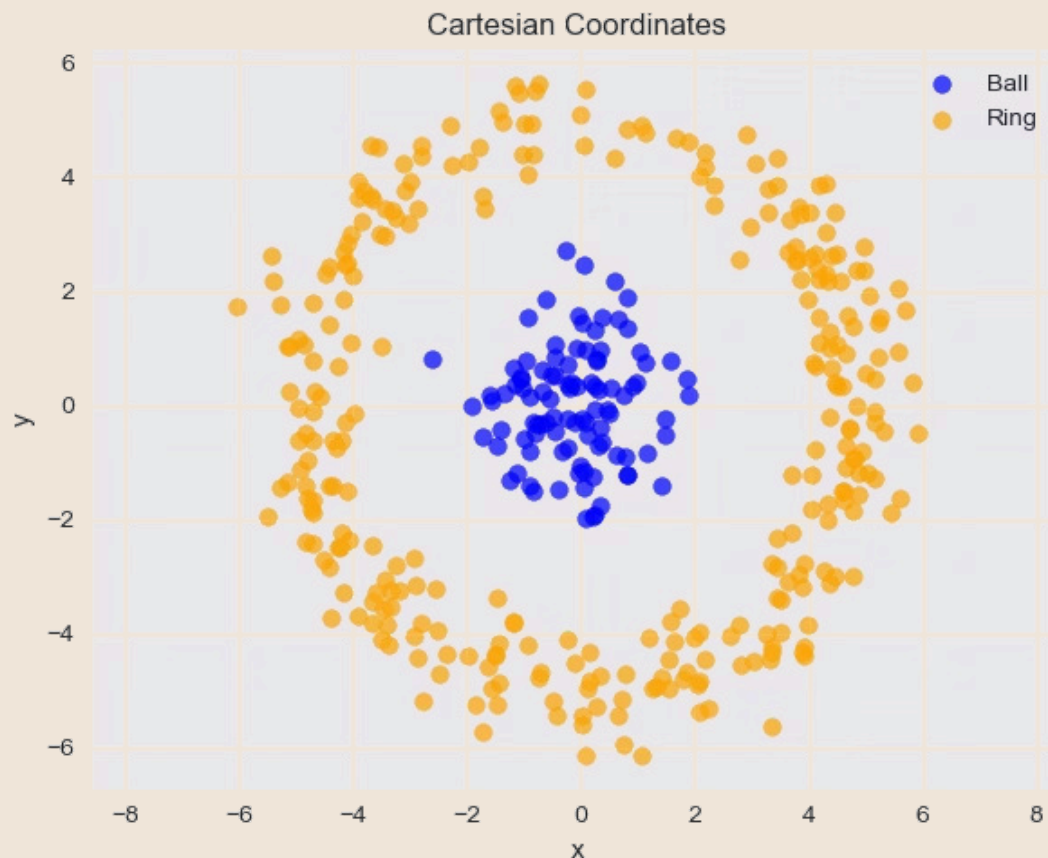
Classification Through Representation

The ball-and-ring problem illustrates how representation transforms classification difficulty. In Cartesian space, no linear boundary can separate the classes. In polar coordinates, a simple threshold on radius solves the problem perfectly.

1

Cartesian Space

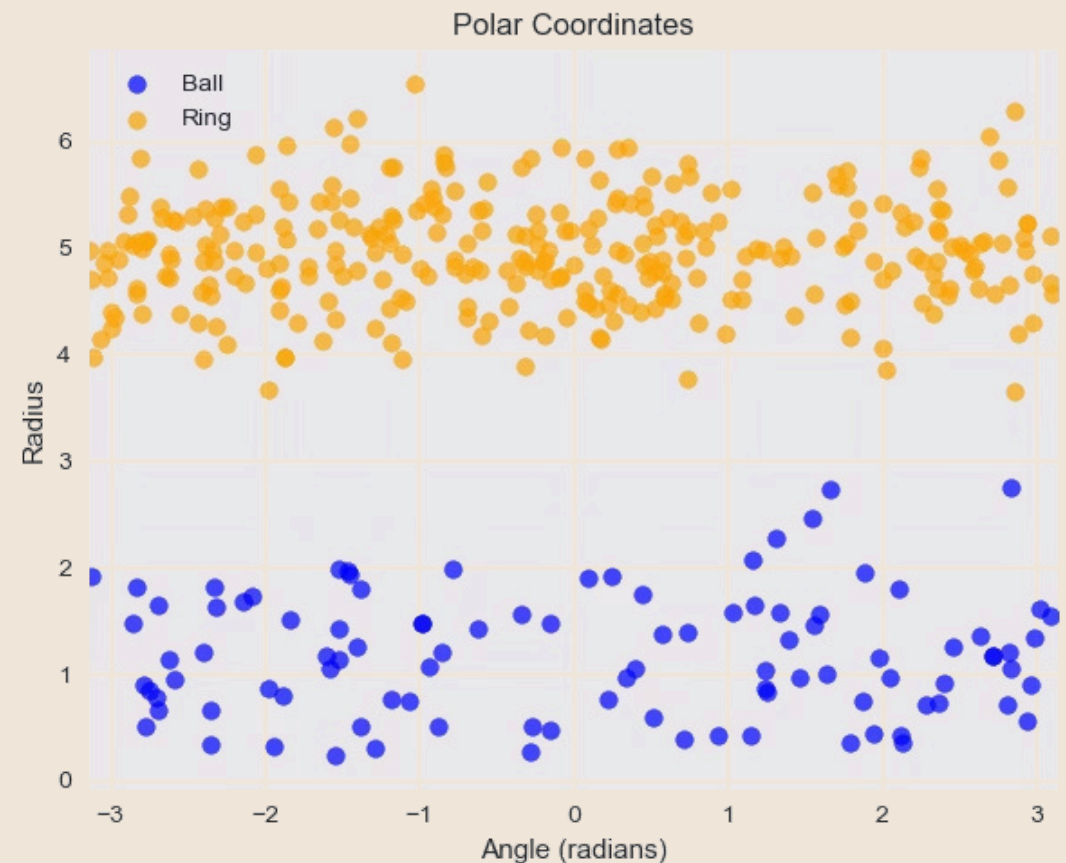
Central ball surrounded by outer ring - impossible for linear classifiers to separate.



2

Polar Transform

Convert to (r, θ) coordinates using radius and angle calculations.



3

Linear Separation

Classes now separated by simple threshold: $r < 3$ (ball) vs $r > 3$ (ring).

From Hand-Crafted to Learned Features

Traditional Approach

Hand-craft feature functions using **human intuition and domain knowledge**. Works for simple cases like polar coordinates, but fails for complex data.

- Requires expert knowledge
- Limited to interpretable features
- Doesn't scale to complex data

Modern Deep Learning

Learn representation functions directly from data using neural networks. Models discover optimal features automatically.

- No manual feature engineering
- Handles complex patterns
- Scales to images, text, audio

 **The Key Question:** What is the "radius" of a cat picture? What is the "angle" of a legal document? For complex real-world data, we can't manually discover perfect features - we must learn them.

Black Box Feature Extractors

Modern machine learning uses massive pre-trained models as powerful feature extractors. Feed in raw data, get out rich numerical representations that capture semantic meaning.



Text Models

BERT, GPT, Sentence Transformers convert sentences into 768-dimensional vectors capturing semantic meaning. Libraries like Hugging Face transformers provide easy access.



Image Models

ResNet, Vision Transformers (ViT) transform raw images into 2048-dimensional vectors representing visual content, objects, and scenes.



Audio Models

MFCCs, Wav2Vec2, Audio Transformers convert raw waveforms into feature vectors representing acoustic content, speaker identity, or spoken words.

DLFeat: Deep Learning Feature Extraction Library

📖 README ⚖️ MIT license



DLFeat: Deep Learning Feature Extraction Library



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DLFeat is a Python library designed for easy and modular feature extraction from various data modalities including images, videos, audio, and text. It leverages powerful pre-trained models from libraries like PyTorch, torchvision, Transformers, Sentence-Transformers, and TIMM. The goal is to provide a "black box" tool suitable for educational and research purposes, allowing users to quickly extract meaningful features for data analysis tasks without needing to delve into the complexities of each model's architecture or training.



Image Models

ResNet, ViT, DINOv2 for visual feature extraction



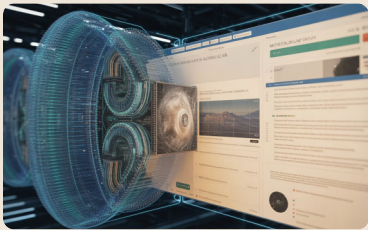
Video Models

Video Swin Transformer, R2Plus1D for temporal features



Audio Models

Wav2Vec2 for acoustic and speech features

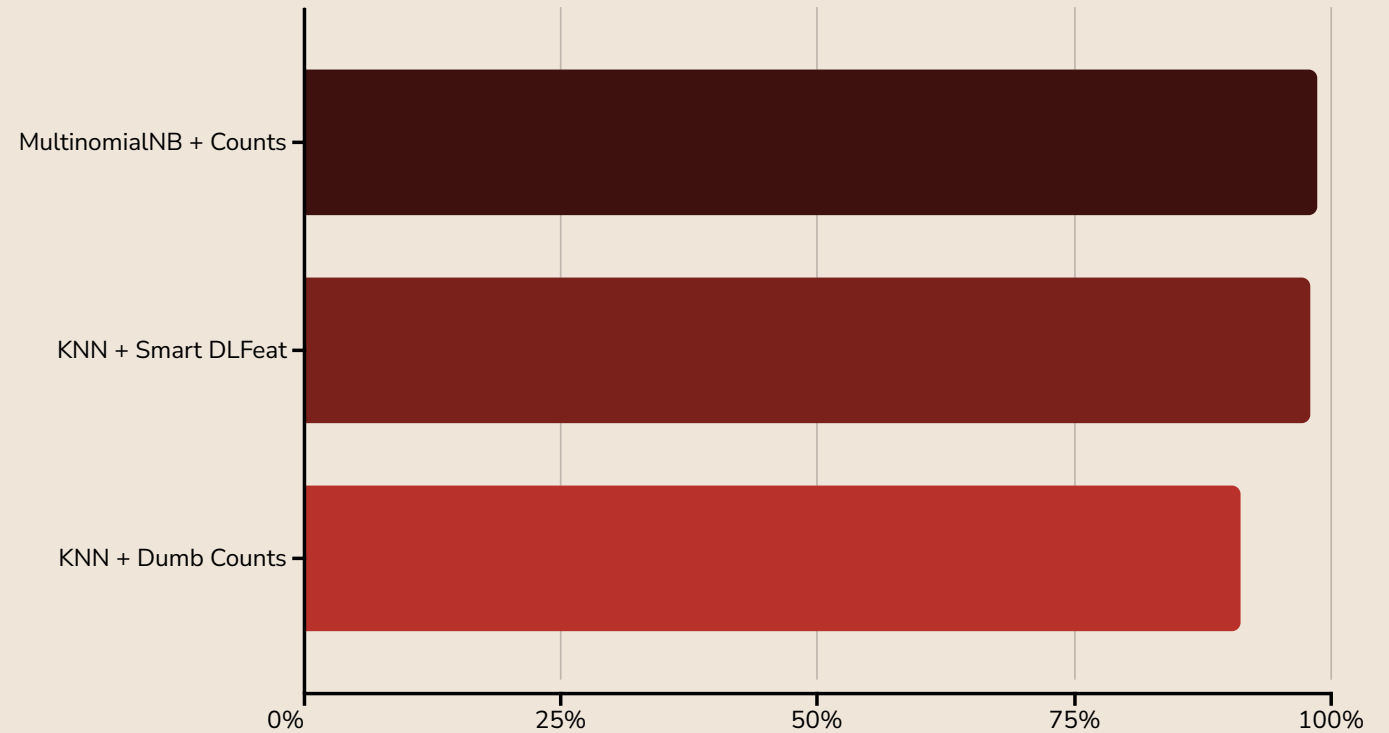


Multimodal Models

CLIP for joint image-text understanding

[Explore DLFeat Documentation](#) [Github Repository](#) [Model Zoo](#)

Laboratory: Feature Extraction and Classification



Conclusions and Next Steps



We Have Explored:

- Feature Representation
- Norms
- Metrics
- Feature Extraction

📅 In the next lectures, we will look at clustering

References

- [https://en.wikipedia.org/wiki/Norm_\(mathematics\)](https://en.wikipedia.org/wiki/Norm_(mathematics)).
- https://en.wikipedia.org/wiki/Lp_space
- https://en.wikipedia.org/wiki/Metric_space
- https://en.wikipedia.org/wiki/Cosine_similarity.
- Parts of Chapter 2 of [1]

[1] Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT press, 2016.